

A Calibrated Separation of the Kontorovich–Lagarias and Volkov Exponent Predictions for the $5x + 1$ Reverse Tree

Renato Augusto Tavares  
Universidade Federal de Goiás
<https://orcid.org/0009-0002-0196-3311>

August 10, 2026

Abstract

Kontorovich and Lagarias predict a counting exponent $\eta \approx 0.650919$ for the $5x + 1$ reverse tree, where $N_u(x) = x^{\eta+o(1)}$ for the accelerated map used here. A competing model due to Volkov predicts $\eta^* \approx 0.678$. The two authors write that Volkov’s own data seems insufficient to discriminate between the values.

We enumerate the reverse tree exactly and extrapolate the truncation bias with Aitken’s Δ^2 . The fixed-window reading is 0.639; it excludes neither prediction. Applied to a branching process built to a known pressure-equation root, the same estimator under-reads by 0.038, more than the gap $\Delta = 0.027$ between the two disputed values, so a raw reading settles nothing.

We build synthetic controls that share the arithmetic tree’s branching law: two also share, via an exact sibling congruence, its sibling spacing, drawing the branch residue instead of the true one; a third draws every branch label independently, with no such congruence. Three such controls built to exponent 0.650919 read within 0.0035 of that value and within 0.0037 of each other. The arithmetic tree reads 0.64926, below the two congruence-matched controls and inside the wider band set by the third. A control built to exponent 0.678 reads 0.0282 away, more than seven widths of that band outside it.

The comparison is a measurement, not a proof, calibrated against controls of known exponent. It targets the disputed exponent value; whether it bears on Volkov’s branching model itself is a separate question.

1 Introduction

Kontorovich and Lagarias [1] study the $3x+1$ and $5x+1$ Collatz-type problems through the counting statistics of the associated reverse tree: for a fixed root u , the number $N_u(x)$ of integers up to x that eventually map to u under the accelerated map satisfies $N_u(x) = x^{\eta+o(1)}$ for a conjectural exponent η . Their stochastic model predicts $\eta_{5,BP} \approx 0.650919$ for $q = 5$ (their Theorem 8.10, subscript BP for branching process, in their own notation). A competing branching model due to Volkov [3], discussed in the same reference, grows a complete binary tree under a different encoding of the iterates and predicts $\eta_{5,BP}^* \approx 0.678$, a gap of $\Delta = 0.027$. Volkov’s model counts a related but different quantity, $Q(x)$: non-divergent trajectories entering some cycle. Kontorovich and Lagarias note $\pi_5(x) \leq Q(x)$, with the two orders of growth only conjectured to agree. Kontorovich and Lagarias write that “the empirical data Volkov presents seems insufficient to discriminate between these two predicted exponents,” and separately note “some controversy concerning the conjectured value of the constant.”

2 Setup: the reverse tree of the accelerated $qx + 1$ map

Fix an odd integer $q \geq 3$. The *accelerated* $qx + 1$ map is

$$T_q(n) = \frac{qn + 1}{2^{v_2(qn+1)}}, \quad n \text{ odd},$$

where v_2 is the 2-adic valuation. The *reverse tree* rooted at an odd integer u coprime to q is built by, at each odd node v , adjoining a child $w_a := (2^a v - 1)/q$ for every $a \geq 1$ such that $2^a v \equiv 1 \pmod{q}$; such w_a is automatically an odd integer whenever it is an integer. Writing $d := \text{ord}_q(2)$, admissibility of a for a given parent v depends only on $v \pmod{q}$: if some $a_0(v) \in \{1, \dots, d\}$ satisfies $2^{a_0(v)} v \equiv 1 \pmod{q}$, it is unique, and the admissible exponents are exactly $a \equiv a_0(v) \pmod{d}$; nodes whose residue admits no such a_0 are sterile. At $q = 5$, $d = 4$, every nonzero residue admits an a_0 , and only $v \equiv 0 \pmod{5}$ is sterile.

Every node has exactly one parent under T_q , so a given integer reaches u , if at all, along a single path. We count them:

$$N_u(x) := \#\{n \leq x : n \text{ odd and } T_q^k(n) = u \text{ for some } k \geq 0\}, \quad N_u(x) = x^{\eta+o(1)}.$$

Remark 2.1. Kontorovich and Lagarias state Theorem 8.10 for the un-accelerated map, counting every integer reaching u under $n \mapsto (qn + 1)/2$ for n odd, $n/2$ for n even. Collapsing the 2-adic halving chain between consecutive odd values changes the count by a factor bounded uniformly in x , so $N_u(x)$ above shares the same exponent η with their un-accelerated counting function. That η exists and does not depend on u is itself their Conjecture 7.2.

Lemma 2.2 (Sibling congruence). *Let $q \geq 3$ be odd, $d = \text{ord}_q(2)$, and v odd with $2^a v \equiv 1 \pmod{q}$ solvable. Write the children of v , in order of increasing admissible exponent, as $w_{(j)} := w_{a_0+jd} = (2^{a_0+jd} v - 1)/q$ for $j \geq 0$, where $a_0 = a_0(v)$. Then*

$$w_{(j+1)} = 2^d w_{(j)} + \frac{2^d - 1}{q} \quad (\text{exact identity in } \mathbb{Z}),$$

and consequently $w_{(j+1)} \equiv w_{(j)} + c \pmod{q}$, with $c := ((2^d - 1)/q) \pmod{q}$.

Proof. Direct substitution: $2^d w_{(j)} + (2^d - 1)/q = (2^{a_0+(j+1)d} v - 2^d + 2^d - 1)/q = w_{(j+1)}$. The congruence follows from $2^d \equiv 1 \pmod{q}$. $\square$

When q is prime, $c = 0$ would force $q^2 \mid 2^d - 1$, the base-2 Wieferich condition; the only known Wieferich primes are 1093 and 3511, so $c \neq 0$ for every q of interest here, and consecutive siblings' residues mod q advance by a fixed nonzero step. At $q = 5$, $d = 4$, $c = 3$. A real denominator q_{val} can replace q in two facts: dropping the additive constant, negligible once $w_{(j)}$ is large, leaves consecutive siblings at multiplicative distance $w_{(j+1)} \approx 2^d w_{(j)}$, independent of q ; and every parent-to-child step itself carries the denominator, $w_a \approx 2^a v/q$, also dropping the additive constant. The residue step c stays integer, mod the original q . §5 uses both facts to build a branching process with the arithmetic tree's sibling spacing but a tunable exponent.

3 Method

We enumerate the exact reverse tree of T_5 from 300 roots: odd integers coprime to 5, sampled uniformly below 10^4 and outside every cycle of T_5 , so the fixed measurement window, 10^5 through

10^8 , lies entirely beyond them. A single depth-first pass tracks the running path-maximum at every node, giving simultaneous counts at truncation buffers 10^9 through 10^{13} ; this agrees, count for count, with an independent buffer-by-buffer implementation on five of the 300 roots. A confidence interval built from a single fixed buffer excludes every candidate exponent once the sampling standard error falls below the residual truncation bias; the path-maximum construction extrapolates the buffer dependence directly, without re-running the enumeration once per buffer. The pooled mean slope across the 300 roots rises with truncation, its increments shrinking geometrically:

$$0.60049 (10^9) \rightarrow 0.62387 (10^{10}) \rightarrow 0.63261 (10^{11}) \rightarrow 0.63650 (10^{12}) \rightarrow 0.63801 (10^{13}),$$

by a factor of 0.37 to 0.45 per decade of truncation: the signature of an extrapolable truncation bias.

4 Result: a fixed-window estimator

Aitken’s Δ^2 extrapolation of this sequence to infinite truncation gives

$$\hat{\eta}_5 = 0.639, \quad 95\% \text{ bootstrap interval, fixed window} = [0.633, 0.645]. \quad (1)$$

The bootstrap interval measures variation between sampled roots after the truncation extrapolation; it does not include the remaining bias from the fixed measurement window, and so cannot exclude either asymptotic prediction. That remaining bias is visible directly in a diagnostic panel: on 40 of the 300 roots, at the deepest buffer 10^{13} , the per-decade slope across $10^4 \rightarrow 10^5$ through $10^7 \rightarrow 10^8$ is still rising at the last decade tested, $0.6021 \rightarrow 0.6296 \rightarrow 0.6432 \rightarrow 0.6460$, with no plateau.

Empirical Result 4.1 (Finite-window reading of the counting exponent). *The fixed-window estimator of the $5x + 1$ reverse-tree counting exponent, computed by exact enumeration with Aitken- Δ^2 -extrapolated truncation correction, is 0.639 (bootstrap interval $[0.633, 0.645]$). The sequence of local slopes rises monotonically across the available windows, still increasing at the last one tested; §5 extrapolates a related per-decade statistic at far greater truncation depth. On its own, this experiment does not provide a calibrated confidence interval for the asymptotic exponent and does not exclude the Volkov prediction.*

5 A calibrated comparison

The estimator behind Empirical Result 4.1 is biased. An *unconstrained construction* shares the arithmetic tree’s offspring law but draws an independent residue at every node; built so its annealed pressure root is the Kontorovich–Lagarias value 0.650919 [2], and read with the same window, truncation, and extrapolation as §4, it gives 0.6131. That is an under-read of 0.038, larger than the gap $\Delta = 0.027$ separating the two disputed values.

The bias falls with depth, and depends on how much a process fluctuates, so it can be calibrated out by comparing readings at a common depth, on a different statistic: not the window-based estimator above, but the single-decade slope $\log_{10}(N(10^{k+1})/N(10^k))$, Aitken-extrapolated over the truncation buffers above 10^{k+1} , read at a checkpoint decade $10^k \rightarrow 10^{k+1}$ deep enough for the bias to have mostly decayed. Two *congruence-matched constructions* share the arithmetic tree’s offspring law and, via Lemma 2.2, its sibling spacing, with a drawn residue at each node: one built to exponent 0.650919, one to 0.678. Replacing the integer denominator q in the parent-to-child step $w_a \approx 2^a v/q$ by a tunable real value q_{val} moves the resulting construction’s exponent α (the smaller root, α_-) to any target solving $q_{\text{val}}^{\alpha_-} = q(2^{\alpha_-} - 1)$, with q itself held fixed at the arithmetic

tree’s own admissibility rule and sterility probability; this is the *real-valued relaxation* of the first congruence-matched construction. The unconstrained construction of the paragraph above, sharing only the offspring law, bounds how much of the calibration depends on the sibling congruence. Every construction below runs on one common set of 300 roots, sampled by the same rule as §3, at checkpoints 10^4 through 10^{10} and truncation buffers 10^9 through 10^{15} , and the arithmetic tree’s reading is checked against them.

Empirical Result 5.1 (Calibrated separation from the Volkov exponent). *At the checkpoint decade $10^9 \rightarrow 10^{10}$, the two constructions built to exponent 0.650919 that share the sibling congruence read 0.65122 (integer recursion) and 0.64981 (its real-valued relaxation), a band 0.00141 wide. The unconstrained construction built to the same exponent reads 0.64751, 0.0034 below target; including it widens the band to 0.0037. The arithmetic tree reads 0.64926, below both congruence-matched constructions and inside the wider band set by the unconstrained one. A construction built to exponent 0.678 reads 0.67748, 0.0282 away from the arithmetic reading, more than seven widths of that wider band. The same comparison at the shallower decade $10^7 \rightarrow 10^8$, on the same checkpoint grid, gives an arithmetic reading of 0.64622 against 0.67200 for the exponent-0.678 construction, a gap of 0.0258: the separation grows with depth. This measurement targets the exponent value 0.678; Volkov’s branching model itself uses a different encoding of the iterates and is not implemented here.*

Pushed further on the arithmetic tree alone, to checkpoints 10^{12} and buffers 10^{17} , past the point where the congruence-matched constructions develop a heavy tail that stalls their own extrapolation, the slope keeps closing on 0.650919: 0.6465 at $10^7 \rightarrow 10^8$, 0.6487 at $10^8 \rightarrow 10^9$, 0.6490 at $10^9 \rightarrow 10^{10}$, then 0.6506 and 0.6505 at $10^{10} \rightarrow 10^{11}$ and $10^{11} \rightarrow 10^{12}$, the last two within 0.0005 of the prediction. This deeper grid’s 0.6490 at $10^9 \rightarrow 10^{10}$ is close to, but computed on different checkpoint and buffer values than, the 0.64926 of Empirical Result 5.1. The bootstrap bands on these five readings cover root resampling only; a separate truncation term, at most 0.002 and falling to at most 0.0004 from $10^8 \rightarrow 10^9$ onward, is a systematic addressed next.

6 Discussion

Two systematics bound Empirical Result 5.1. The congruence-matched construction and its real-valued relaxation, both built to exponent 0.650919, agree to 0.0035 over six seeds on the window estimator, bounding any implementation systematic between the two recursions at about 0.004, comparable to the wider band width itself. Separately, the construction built to exponent 0.678 fluctuates more than the one built to 0.650919 and, on the same window estimator, under-reads its own target by more: 0.019 against 0.011. More fluctuation means more bias, so this pulls the exponent-0.678 reading down, toward the arithmetic reading; the separation survives that direction of bias by construction. The implementation systematic alone is comparable to the 0.0037 wider band width used to read the comparison, and well below the 0.0282 separation itself.

The 0.038 bias of the opening measurement in §5 and the residual bias at the calibration depth, under 0.004 for every construction there, are the same estimator at different depths. The bias falls by roughly an order of magnitude between the window used for the first measurement and the deeper checkpoint decade used for the second.

The comparison settles the narrower question of whether the arithmetic tree’s counting exponent is closer to 0.650919 or to 0.678. It does not implement or test Volkov’s branching model directly, which encodes the iterates differently from the matched constructions built here. The unconstrained construction shares its mean offspring intensity with Kontorovich and Lagarias’s own accelerated branching model: each admissible exponent is fertile with probability $1/5$ in both, giving the same

pressure function and the same exponent by the companion paper’s argument for that intensity [2]. The two processes differ in their joint structure: the unconstrained construction draws one residue class per node, while the accelerated Kontorovich–Lagarias model assigns each exponent independently. On the same 300 roots and grid, the gap between the unconstrained construction’s bias (0.038) and the arithmetic tree’s (0.013, presuming the disputed exponent is 0.650919) measures how much less the arithmetic tree fluctuates than a process sharing its intensity, consistent with the arithmetic tree fluctuating less than every construction in §5 and reading below both constructions built to 0.650919 at the calibration depth. The deep arithmetic-only run sits systematically at or just below 0.650919, a pattern the data here cannot yet distinguish from a slowly converging extrapolation.

Data Availability Statement

Code and data reproducing every claim in this paper, organized by section with a README per subfolder, are at <https://github.com/faculdade/collatz-kl-volkov>.

Acknowledgments

AI assistance (Claude, Anthropic) was used for textual review and translation.

References

- [1] A. V. Kontorovich and J. C. Lagarias, *Stochastic models for the $3x + 1$ and $5x + 1$ problems*, arXiv:0910.1944 (2009); published in: *The Ultimate Challenge: the $3x + 1$ Problem* (J. C. Lagarias, ed.), American Mathematical Society, 2010. Theorem 8.10 and the following Remark are cited from the arXiv version.
- [2] R. A. Tavares, *A closed-form pressure equation for the accelerated $qx + 1$ branching process*, companion paper, in preparation.
- [3] S. Volkov, *A probabilistic model for the $5x + 1$ problem and related maps*, Stochastic Processes and their Applications **116**, no. 4 (2006), 662–674.